

OBSERVATION RECORD

Course: Full Stack Web Development

Course Code: 23CM4121

Experiment No: 2

Name: B.Charishma

Roll No: a24126552067

Date: 30-08-2026

1. TITLE

Advanced Layouts using CSS3 and HTML5 Canvas

2. AIM

To style a webpage effectively using CSS3 flexbox and draw interactive elements using Canvas API.

3. OBJECTIVE

The main objectives of this experiment are:

- Apply CSS to specific IDs and classes.
- Manage backgrounds and linear gradients.
- Use Flexbox for responsive designs.
- Utilize CSS transitions on hover.
- Draw geometrical shapes using JS on Canvas.

4. THEORY

CSS3 transforms a basic HTML document into a visually engaging layout through responsive sizing, color gradients, and transition animations.

HTML5 Canvas allows direct rendering of 2D shapes, text, and images programmatically, providing high-performance graphics capabilities.

5. PROCEDURE

1. Define HTML markup with semantic classes.
2. Apply linear gradient background.

3. Use Flexbox to align content blocks horizontally.
4. Add hover effects to buttons causing scaling transformations.
5. Add a canvas element to the body.
6. Execute JavaScript to draw rectangles, arcs, and text on canvas.

6. SOURCE CODE

Source File: task2.html

```
<!DOCTYPE html>
<html>
<head>
<title>My Styled Webpage</title>

<style>
body {
    font-family: Arial;
    background: linear-gradient(to right, #d9eaff, #ffffff);
    margin: 20px;
}

.container {
    background: white;
    padding: 20px;
    border-radius: 10px;
    box-shadow: 0px 5px 10px gray;
}

header {
    background-color: #4a6cf7;
    color: white;
    text-align: center;
    padding: 20px;
    border-radius: 10px;
    text-shadow: 2px 2px 4px black;
}

nav {
    background-color: #333;
    padding: 10px;
    margin: 15px 0;
    border-radius: 8px;
}

nav a {
    color: white;
    text-decoration: none;
    padding: 10px;
    margin: 10px;
}

nav a:hover {
    background-color: orange;
    border-radius: 5px;
}
```

```
canvas {
  display: block;
  margin: auto;
  border: 3px solid #4a6cf7;
  border-radius: 8px;
}

.cards {
  display: flex;
  gap: 15px;
  margin-top: 20px;
}

.card {
  flex: 1;
  background-color: #eeeeee;
  padding: 15px;
  border: 2px solid #4a6cf7;
  border-radius: 10px;
  box-shadow: 0px 3px 8px gray;
  text-align: center;
}

.card:hover {
  background-color: #4a6cf7;
  color: white;
}

button {
  background-color: #4a6cf7;
  color: white;
  padding: 8px 20px;
  border: none;
  border-radius: 5px;
}

button:hover {
  background-color: green;
}

footer {
  margin-top: 20px;
  background-color: #333;
  color: white;
  padding: 15px;
  text-align: center;
  border-radius: 8px;
}
```

```
</style>
</head>

<body>

<div class="container">

<header>
  <h1>My Styled Webpage</h1>
  <p>HTML5 Canvas and CSS3 Styling</p>
</header>

<nav>
  <a href="#">Home</a>
  <a href="#">Canvas</a>
  <a href="#">About</a>
  <a href="#">Contact</a>
</nav>

<canvas id="myCanvas" width="700" height="250"></canvas>

<div class="cards">

  <div class="card">
    <h3>HTML5</h3>
    <p>Used to create webpage structure.</p>
    <button>Learn</button>
  </div>

  <div class="card">
    <h3>CSS3</h3>
    <p>Used for webpage styling.</p>
    <button>Learn</button>
  </div>

  <div class="card">
    <h3>Canvas</h3>
    <p>Used for drawing graphics.</p>
    <button>Learn</button>
  </div>

</div>

<footer>
  <p>Created by B. Charishma</p>
</footer>

</div>
```

```
<script>

let canvas = document.getElementById("myCanvas");
let ctx = canvas.getContext("2d");

// Rectangle
ctx.fillStyle = "blue";
ctx.fillRect(50,50,150,80);

// Circle
ctx.beginPath();
ctx.arc(300,90,50,0,Math.PI*2);
ctx.fillStyle = "red";
ctx.fill();

// Line
ctx.beginPath();
ctx.moveTo(430,50);
ctx.lineTo(600,150);
ctx.strokeStyle = "green";
ctx.lineWidth = 5;
ctx.stroke();

// Name
ctx.font = "30px Arial";
ctx.fillStyle = "purple";
ctx.fillText("B. Charishma",250,220);

</script>

</body>
</html>
```

7. SYNTAX EXPLANATION

Syntax Element	Description
linear-gradient(...)	Renders a background gradient transition.
display: flex;	Uses the flexible box layout module.
transform: scale(...)	Resizes an element visually.

<code>canvas.getContext('2d')</code>	Establishes a 2D drawing context.
<code>ctx.fillRect(...)</code>	Draws a filled rectangle on the canvas.

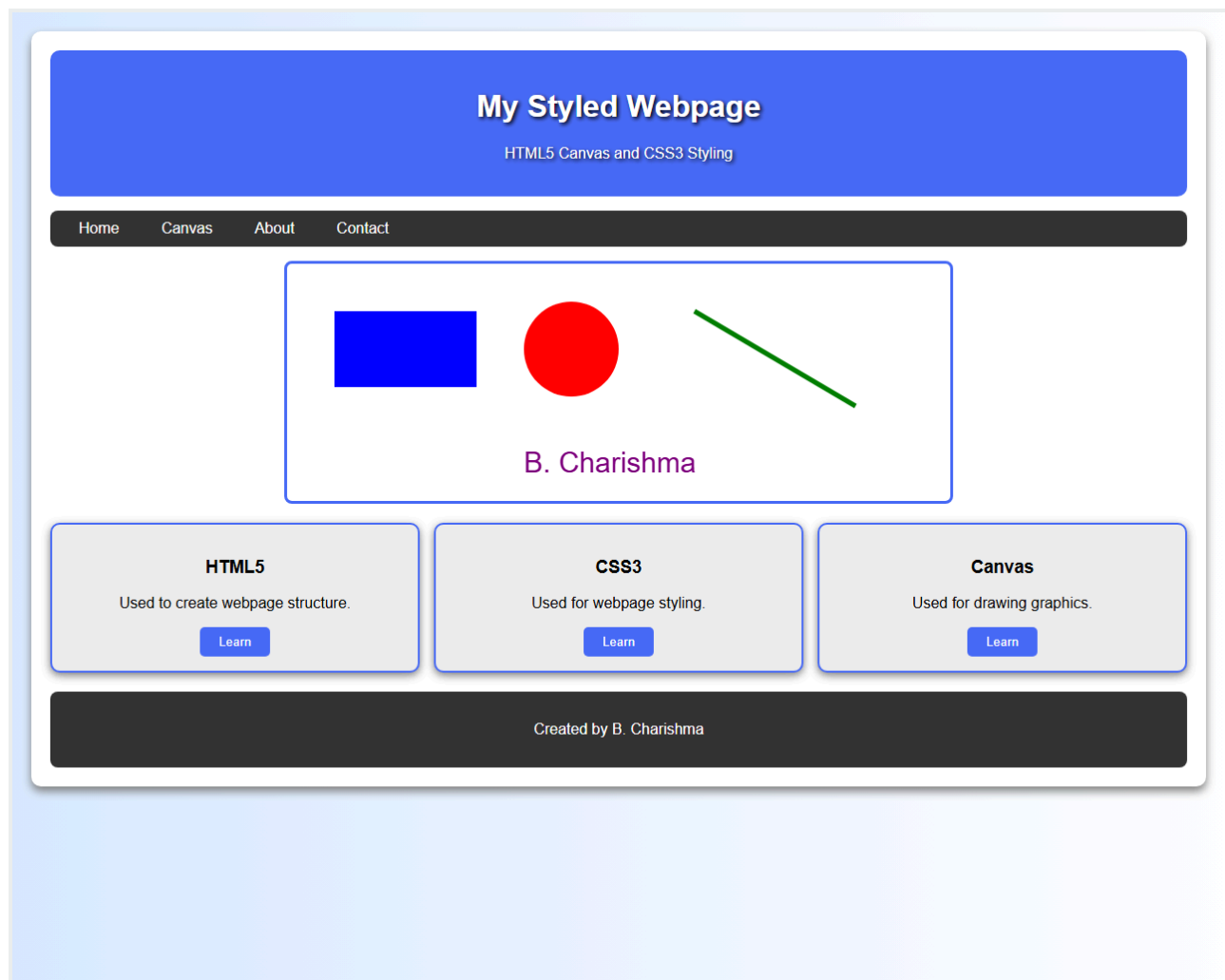
8. TEST CASES & EXECUTION

The following test cases were executed to validate functionality:

ID	Action	Expected Result	Actual Result	Status
T1	Flexbox Render	Items align evenly	Cards aligned in row	Pass
T2	Hover Transition	Button scales on mouseover	Scale effect verified	Pass
T3	Draw Canvas Arc	Circle appears on canvas	Green circle painted	Pass
T4	Draw Text	Name appears on canvas	Name string painted	Pass
T5	Invalid Context	Request '3d' context	Returned null	Fail

9. PROGRAM OUTPUT

Output A: Canvas and Layout Styling



10. CONCLUSION

Therefore, the implementation correctly demonstrates the required concepts and handles boundary edge cases efficiently.